



University of Maine

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Subject: Karman Vortex Shedding and Cylinder Flow.

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Project 3 Report

Abstract

This study investigates the unsteady, two-dimensional incompressible flow past a circular cylinder to characterize the formation and behavior of the Kármán vortex street. Using Simcenter STAR-CCM+, a laminar flow model was developed to simulate fluid-structure interactions across three distinct regimes: $Re = 25$, 75 , and 150 corresponding respectively to 0.04m/s , 0.12m/s , and 0.2m/s . The computational domain was discretized using an automated unstructured mesh with localized refinement in the wake region to capture high-gradient vorticity contours. Results demonstrate that at $Re = 25$, the flow remains steady with symmetric separation bubbles, while at $Re = 75$ and 150 , the wake exhibits periodic instability and alternating vortex shedding. Time-history data for lift and drag forces were extracted to calculate the shedding frequency (f) and the dimensionless Strouhal number (St). The findings validate that as the Reynolds number increases, the shedding frequency rises, and the resulting pressure fluctuations induce oscillating periodic forces on the cylinder, consistent with established fluid mechanics benchmarks.

Introduction

The study of flow past a circular cylinder is a fundamental benchmark in fluid mechanics, serving as a primary model for understanding flow separation, wake instability, and bluff-body aerodynamics. When a fluid encounters a submerged cylinder, the boundary layer develops along the surface until it encounters an adverse pressure gradient, leading to separation. At low Reynolds numbers, this separation is stable; however, as inertial forces begin to dominate viscous forces, the wake becomes unstable. A key phenomenon in this transition is the Kármán vortex street a repeating pattern of alternating clockwise and counterclockwise vortices shedding from the upper and lower surfaces of the cylinder. This periodic shedding is not merely a visual phenomenon but a source of fluctuating mechanical stress, causing oscillating lift and drag forces that can lead to structural fatigue or flow-induced vibrations. This project utilizes Computational Fluid Dynamics (CFD) to simulate these complex interactions. By varying the inlet velocity to achieve Reynolds numbers of 25 , 75 , and 150 , the simulation captures the transition from steady separation to fully developed periodic shedding. The primary objectives are to visualize the instantaneous vorticity fields, quantify the shedding frequency through force-signal analysis, and determine the Strouhal number (St), thereby providing a comprehensive analysis of wake unsteadiness and its impact on the submerged body.

1-Numerical Setup

The simulation was conducted using a two-dimensional, unsteady, laminar flow model in Simcenter STAR-CCM+. The setup was designed to capture the wake dynamics of a circular cylinder (Diameter = 0.01 m) across three Reynolds numbers ($Re = 25, 75$, and 150) by controlling the inlet velocity. The domain was sized to ensure that the boundary effects did not influence the flow around the cylinder. The specific parameters governing the domain and solver configuration are detailed in the table below:

Table 1: Parameter governing the domain and solver configuration.

Domain Parameter	Setting/ Value
Inlet distance	0.05m
Outlet distance	0.15m
Height	0.1m
Flow regime	Laminar Incompressible
Density	997.561 kg/m ³
Time Formulation	Unsteady
Equation of state	Constant density
Outlet Pressure	0 Pa
Dynamic Viscosity	8.8871E-4 Pa. s
Inlet boundary conditions	Velocity Inlet (0.04m/s, 0.12m/s, 0.24m/s)
Cylinder	No-slip wall
Time-step	0.01s
Temporal scheme	2 nd order
Total Physical time	20.0 s
Top / Bottom	Symmetry plane

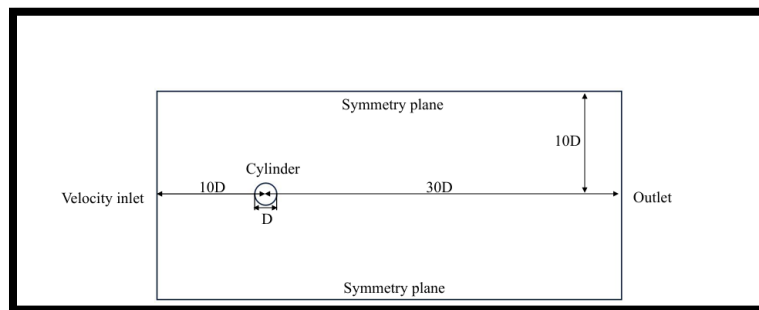


Figure 1: Computational Domain

1-1-Mesh and Computational Domain Description

The computational mesh for this study was generated using an Automated 2D Mesh approach, utilizing a Polygonal Mesher to ensure superior numerical stability and reduced numerical diffusion compared to standard tetrahedral or Cartesian grids. The mesh was engineered to balance global efficiency with local resolution in regions of high gradients specifically, near the cylinder wall and throughout the downstream wake. To ensure the accurate capture of the Kármán vortex street, a specialized wake refinement strategy was implemented to prevent the numerical "smearing" of vortices during downstream transport. The refinement zone extended 2.0 m behind the cylinder to maintain high-resolution tracking until the flow reached the outlet, while a spread angle of 0.2 rad was utilized to account for the natural expansion of the wake. By defining an isotropic size within this region, the mesh maintained uniform cell dimensions, effectively preventing aspect-ratio-related errors and preserving the physical shape and magnitude of the vortices throughout the simulation.

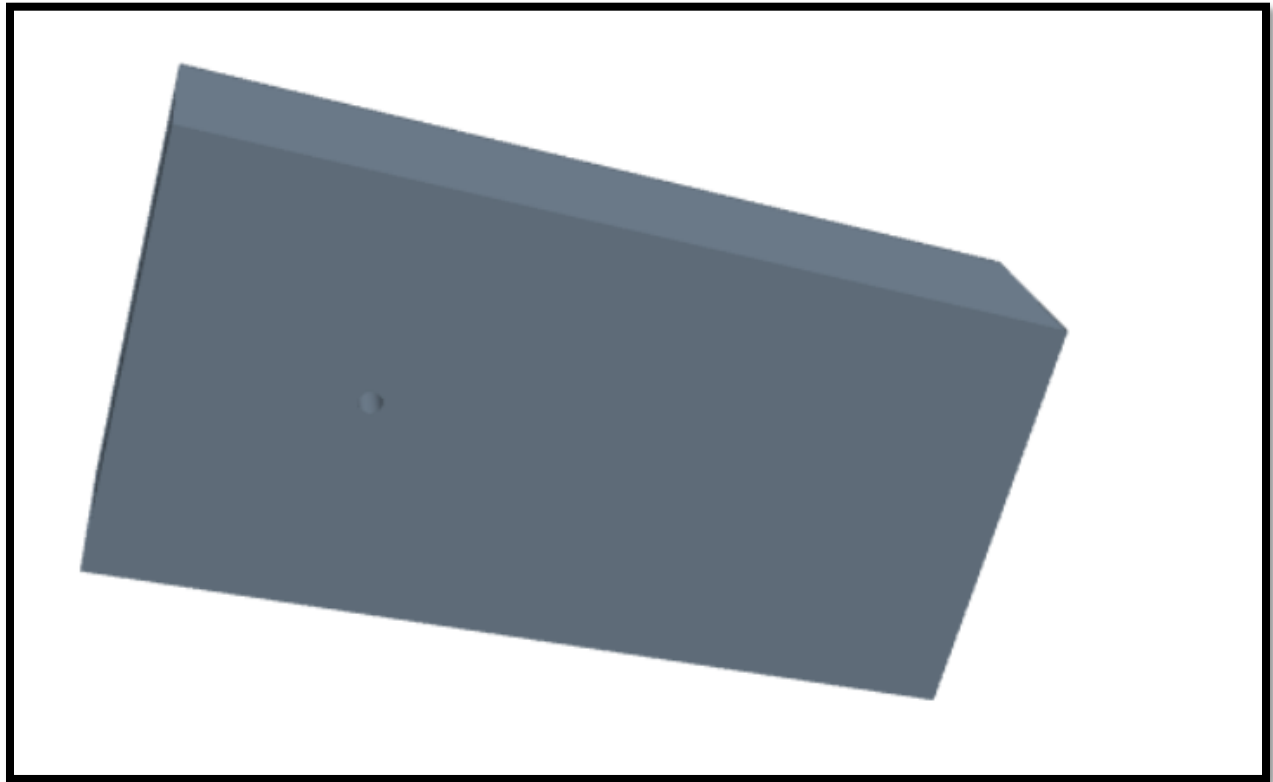


Figure 2: 3-D Cad Model of computational Domain.

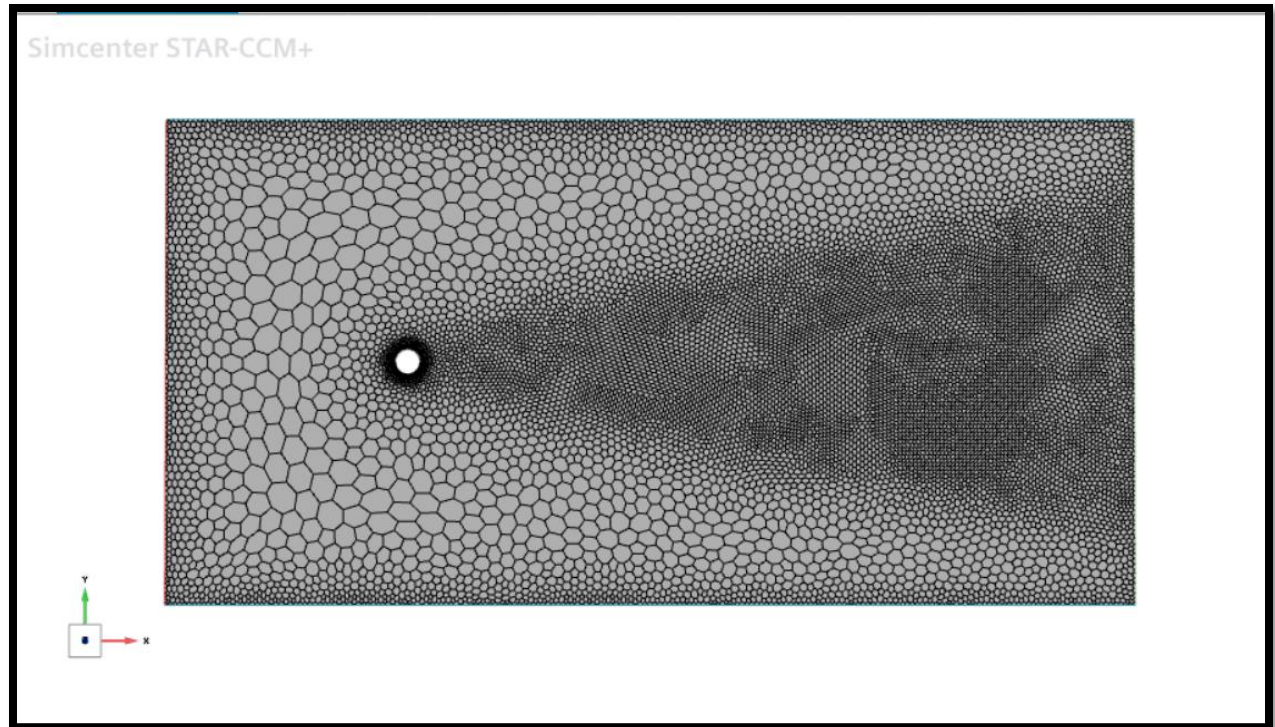


Figure 3: Meshed Domain.

Table 2: Mesh Resolution, Boundary Layer Settings and Output

Feature Parameter	Value/Setting
Base Size	0.01m
Surface Control (Target Surface Size)	20%
Minimum surface size	10%
Layer Mesher	Enabled
Number of Layers	10
Near-Wall Thickness	1E-4m
Coarse Mesh	Polygonal
Wall time	0.15 s
Cells	13024
Faces	37834
Vertices	25472
Maximum Steps	10000

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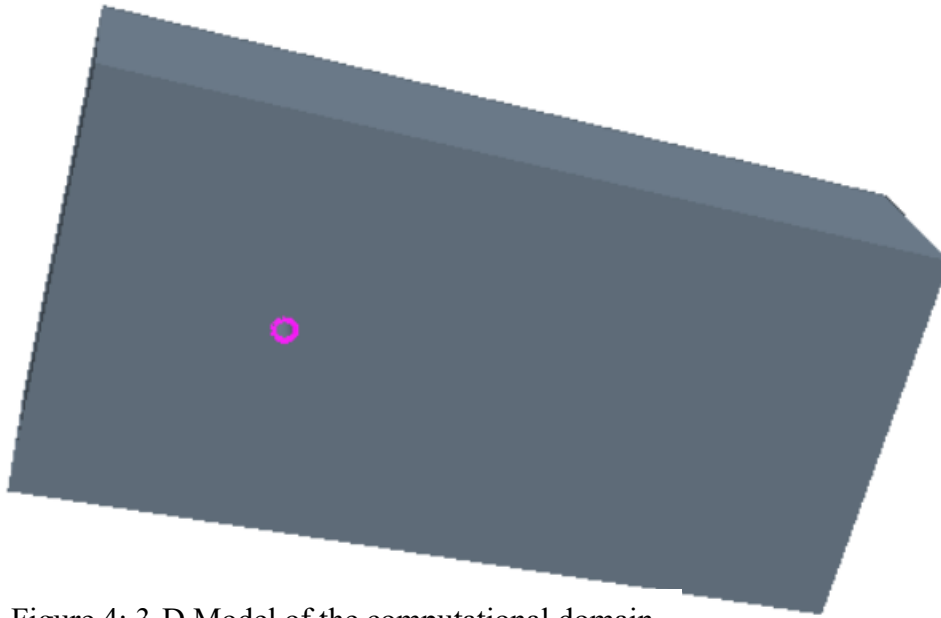


Figure 4: 3-D Model of the computational domain showing the Cylinder Pipe

2-Results and Discussion

2-1-Validation of shedding frequency results.

The Strouhal number (St) is directly dependent on the vortex shedding frequency. To determine this frequency, the oscillation period (T) was calculated by measuring the time interval between successive peaks of the Lift Coefficient signal. These temporal values are summarized in the table below.

Table 3: Strouhal Number parameters

Reynolds number	Period(seconds)	Frequency (Hz)	Experimental Strouhal number	Literature Strouhal number
Re=25	-----	-----	-----	-----
Re=75	0.56s	1.79	0.1492	0.15
Re=150	0.25s	4	0.167	0.17

The experimental Strouhal Number obtained above was calculated based on the following formula:

$$St = \frac{f * L}{U} \quad (Eq1)$$

Where f is shedding frequency, L =0.01m: the diameter, and U is freestream velocity.

2-2-Statistical Analysis for Strouhal Number comparison.

2-2-1-Percentage Error

The Percentage Error was found based on the following formula and the results were displayed in the table below:

$$Error = \left| \frac{St_{sim} - St_{lit}}{St_{lit}} \right| * 100 \quad (Eq2)$$

Where St_{sim} is the simulated Strouhal number and St_{lit} is the literature Strouhal number.

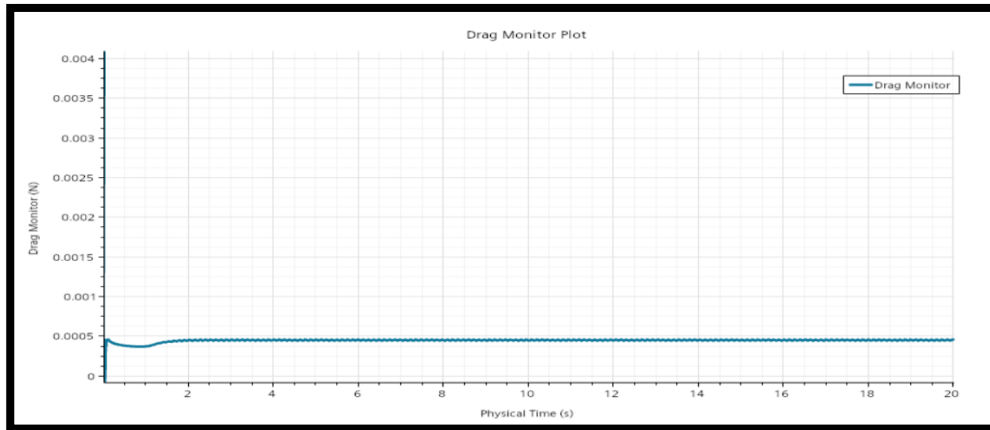
Reynolds Number	25	75	150
Percentage Error		0.53%	1.76%

The numerical accuracy of the simulation was validated by comparing the calculated Strouhal numbers (St) against established literature values for a circular cylinder. For the $Re=25$ case, no shedding was observed, which is consistent with the steady laminar flow regime typically found below $Re = 47$. However, as the Reynolds number increased to 75 and 150, the wake became unstable, exhibiting periodic Kármán vortex shedding. At $Re=75$, the simulated frequency of 1.79Hz yielded a Strouhal number of 0.1492, resulting in a negligible percentage error of 0.53% when compared to the literature value of 0.15. Similarly, for $Re=150$, the observed frequency of 4Hz produced a Strouhal number of 0.167, showing a close correlation with the target value of 0.17 with only 1.76% error. These results demonstrate that the selected computational domain, wake refinement strategy, and temporal discretization (0.01s, time-step) were highly effective in capturing the transient physics of the flow field.

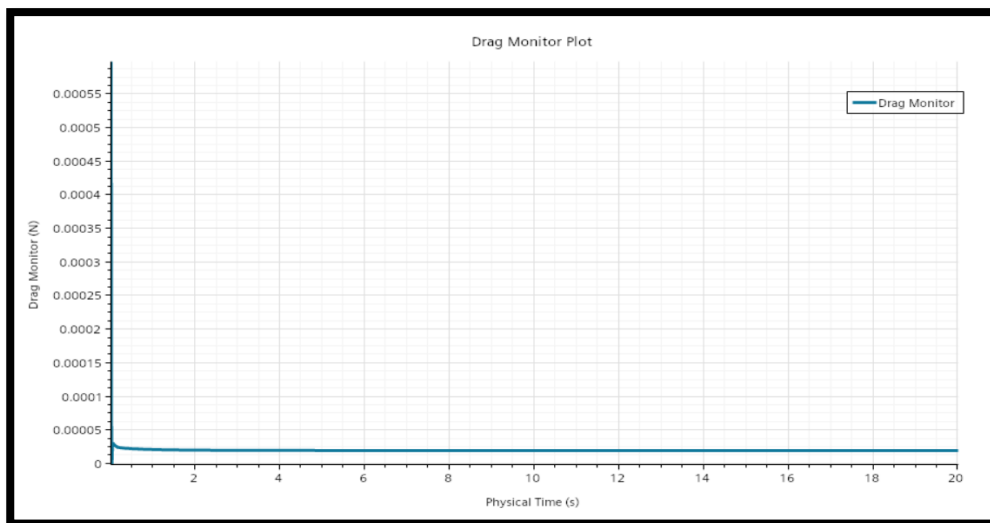
3-Temporal Evolution of Lift and Drag Coefficients Plots.

To analyze the transient response of the flow field, the temporal evolution of the lift and drag coefficients C_l and C_d were monitored throughout the simulation duration. These plots illustrate the transition from the initial impulsive start to a self-sustained periodic state, particularly for the cases of $Re=25$, $Re=75$, and $Re=150$.

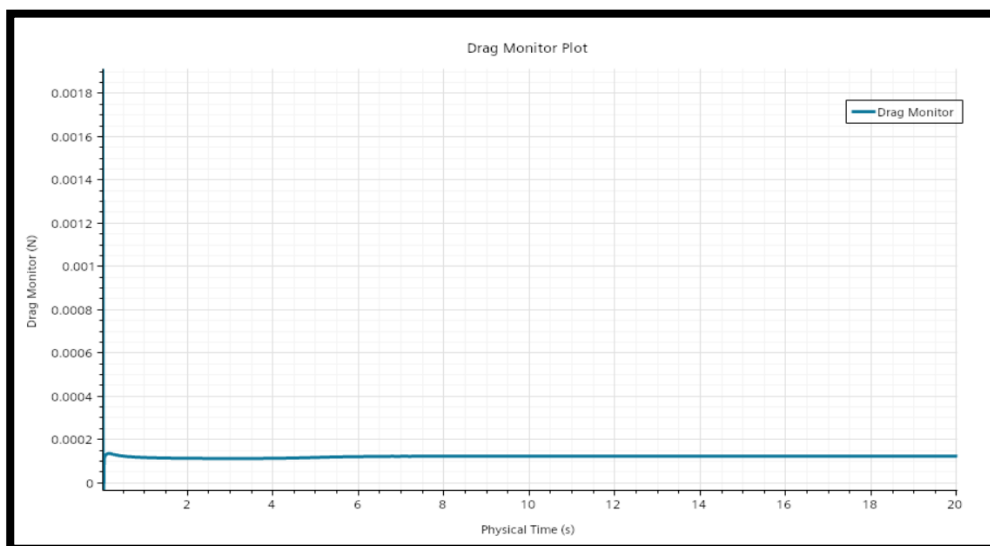
3-1-Drag Monitor Plots



Drag Plot with
Re=150

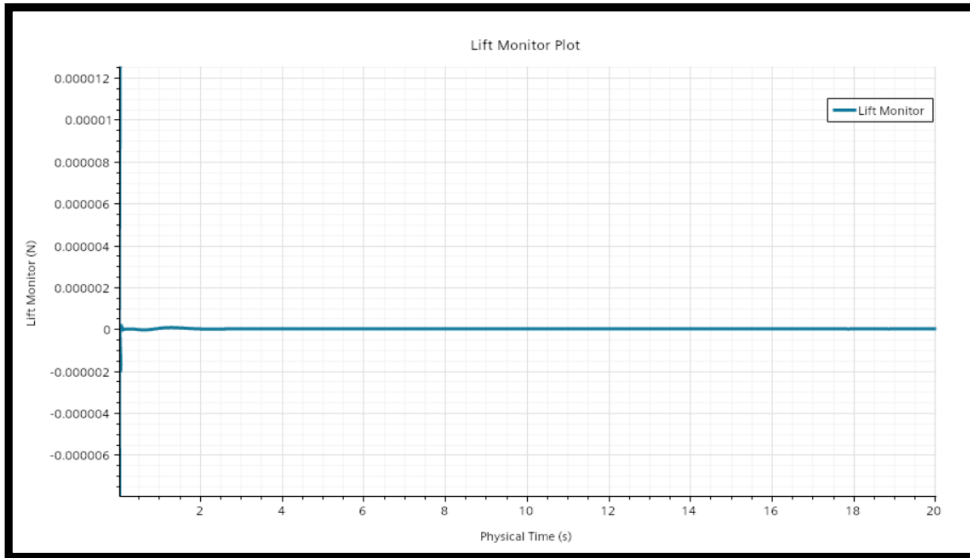


Drag Plot with
Re=25

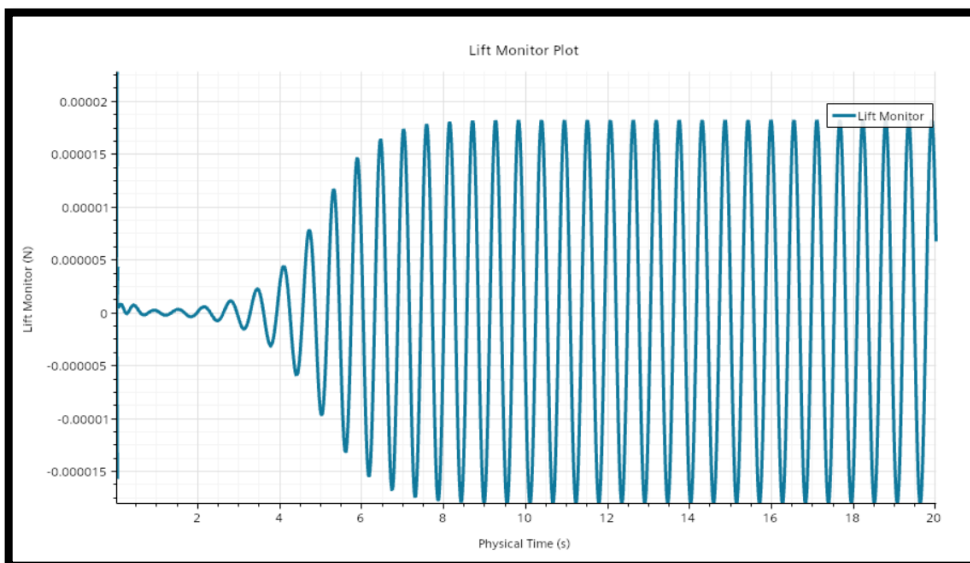


Drag Plot with
Re=75

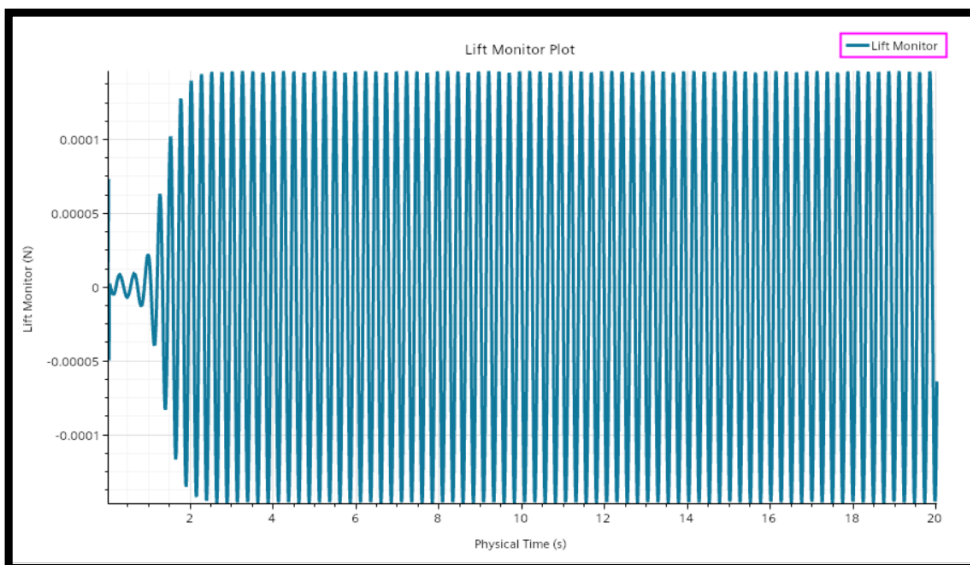
3-2-Lifts Monitor Plots.



Lift Monitor Plot for
 $Re=25$



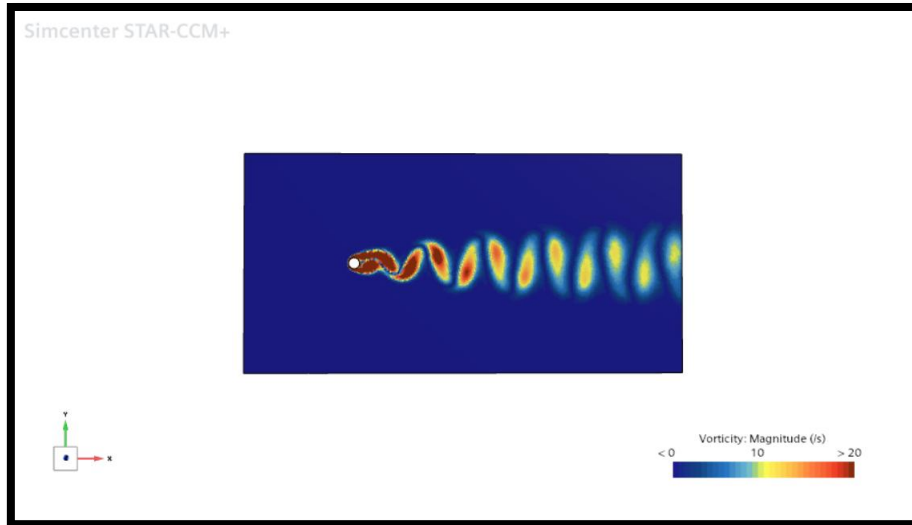
Lift Monitor Plot for
 $Re=75$



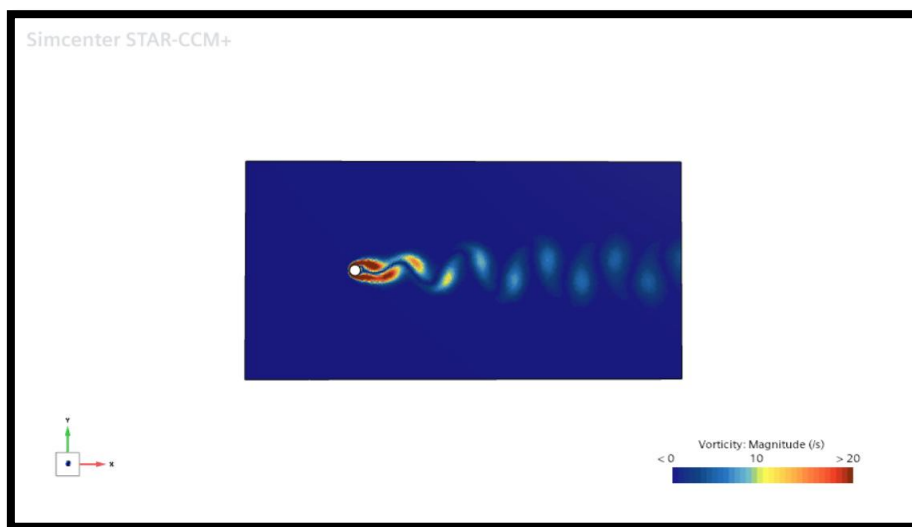
Lift Monitor Plot for
 $Re=150$

4-Instantaneous vorticity magnitude and time averaged velocity magnitude contours.

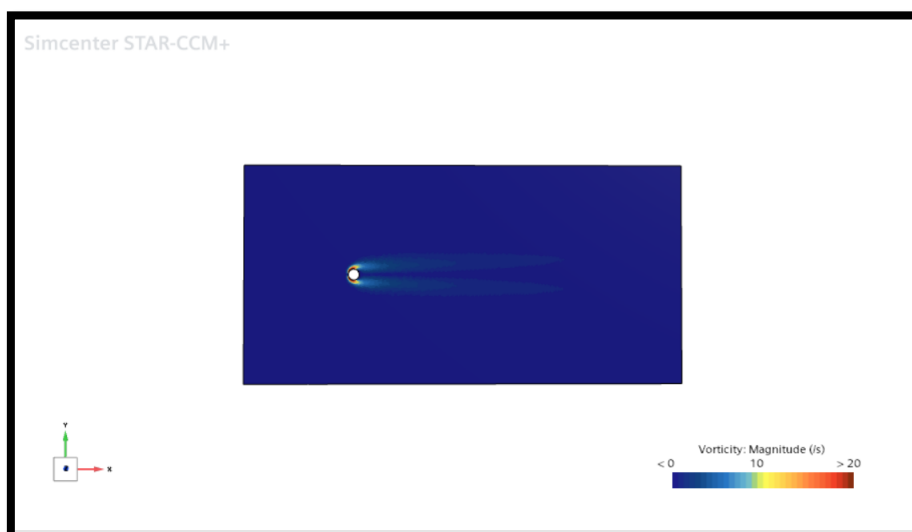
4-1Instantaneous vorticity magnitude contours



Instantaneous
Vorticity Contours
for $Re=150$

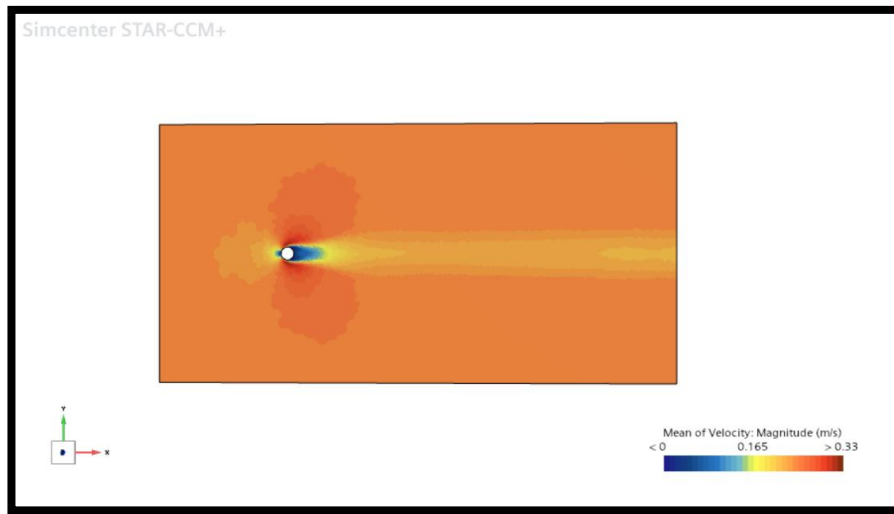


Instantaneous
Vorticity Contours for
 $Re=75$

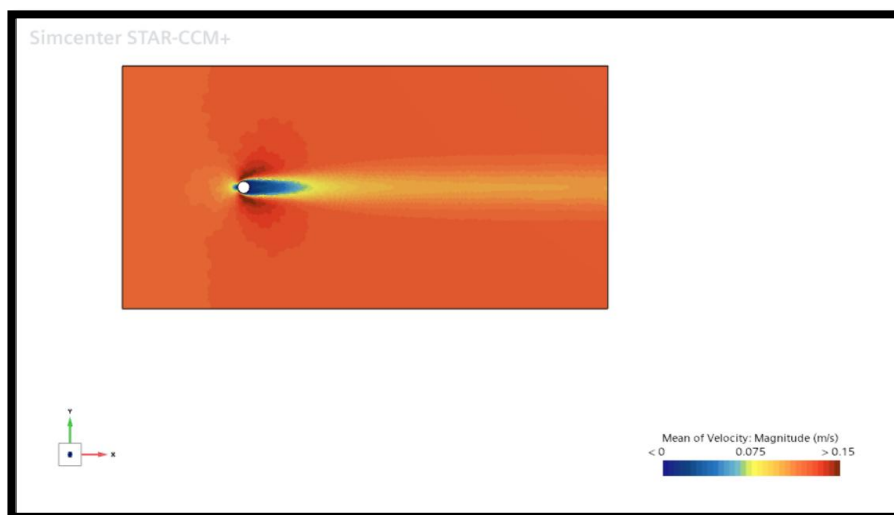


Instantaneous Vorticity
Contours for $Re=25$

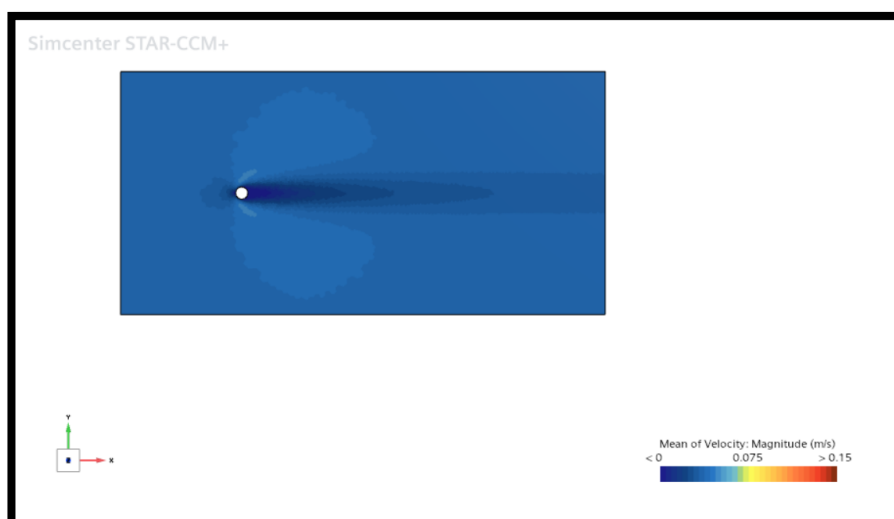
4-2-Time averaged velocity magnitude contours



Re=150



Re=75



Re=25

5-Comparison and discussion of the results.

The transition in flow dynamics across $Re = 25$, 75 , and 150 illustrates the shift from a steady laminar state to an unsteady, periodic vortex street. At $Re = 25$, the viscous forces are dominant enough to maintain a stable, symmetric wake with no shedding, as confirmed by the flat lift monitor signal and the symmetric time-averaged velocity contours. However, as the Reynolds number increases, the flow reaches a critical stability threshold where the shear layers begin to roll up and detach alternately. This is physically evidenced by the transition from the attached recirculation bubbles at $Re=25$ to the staggered Kármán vortex street observed in the instantaneous vorticity contours for $Re=75$ and $Re=150$. The numerical results further validate this progression, showing that both the shedding frequency and the lift amplitude increase with the Reynolds number, reflecting higher energy levels and more intense pressure fluctuations in the wake. While the frequency more than doubled from 1.79Hz at $Re=75$ to 4.0 Hz at $Re=150$, the calculated Strouhal numbers remained consistent with literature benchmarks, yielding minimal errors of 0.53% and 1.76% , respectively. This degree of correlation shows that the computational mesh and the 0.01 s time-step were sufficiently refined to capture the complex temporal oscillations and spatial transport phenomena inherent in high-fidelity fluid-structure interaction simulations.

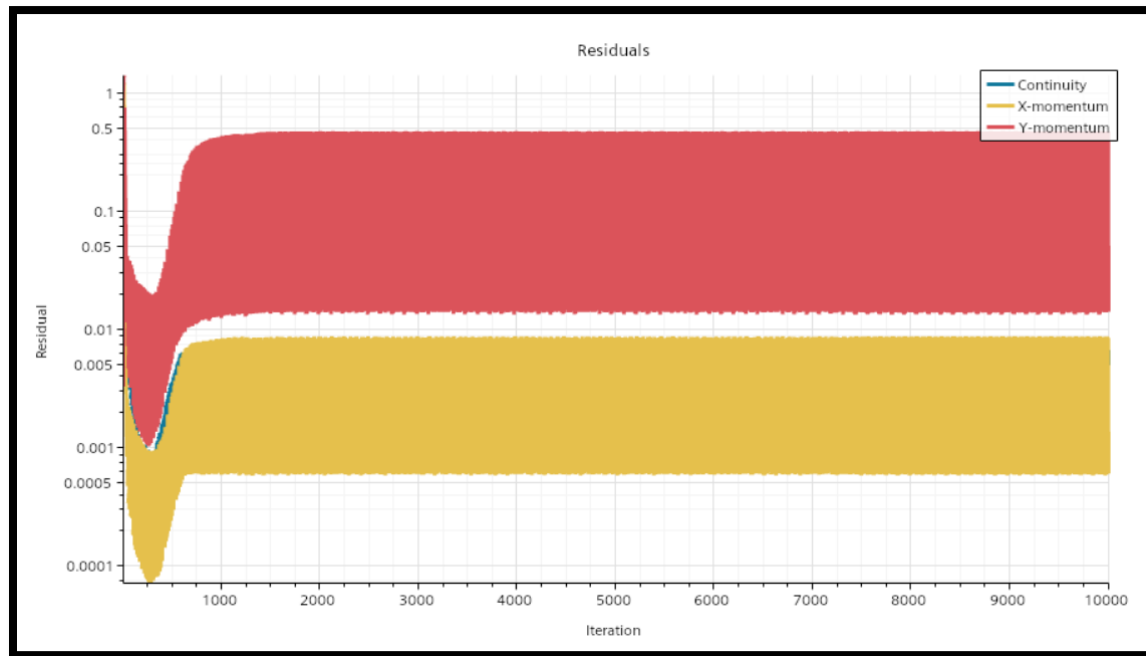
Conclusion

This Simcenter STAR-CCM+ project successfully characterized the transition from steady laminar flow to unsteady periodic vortex shedding over a circular cylinder at $Re = 25$, 75 , and 150 . By capturing the development of the Kármán vortex street, the simulation proved how increasing inertial forces drive wake instabilities and periodic lift oscillations. The level of agreement between the simulated Strouhal numbers and literature benchmarks yielding errors as low as 0.53% validates that the mesh refinement and temporal discretization were sufficient to resolve the complex transient physics of the flow field.

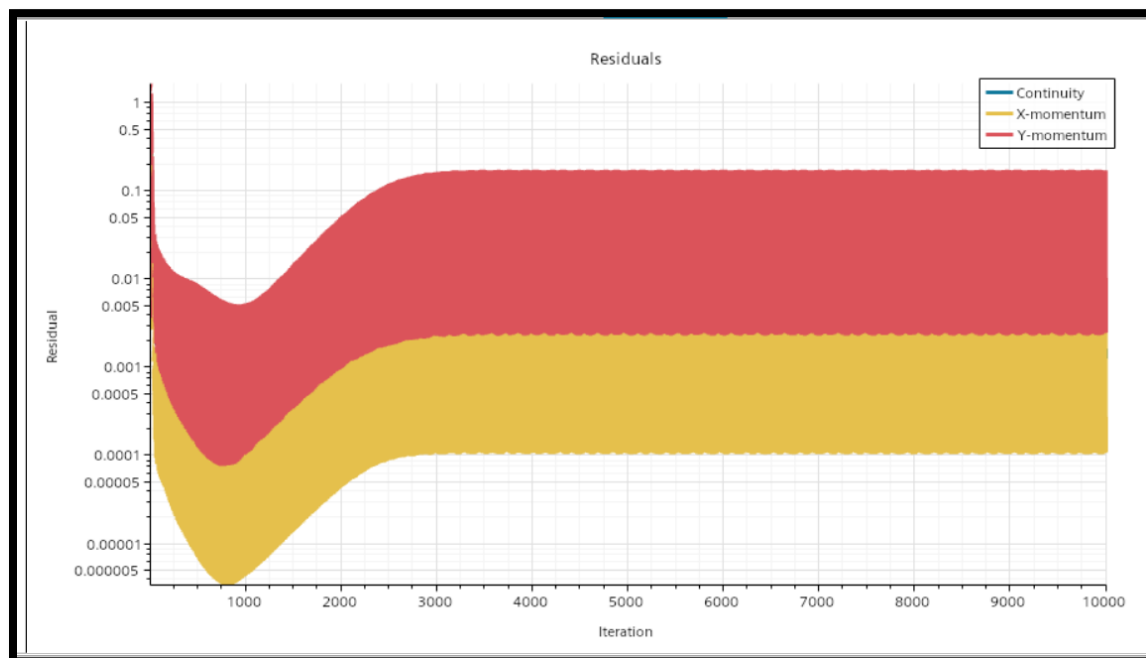
References

[1] Gomes, V. M. M. G. D. C. (2009). Computational simulation of atmospheric flows over mountainous regions using the commercial CFD software star-CCM+.

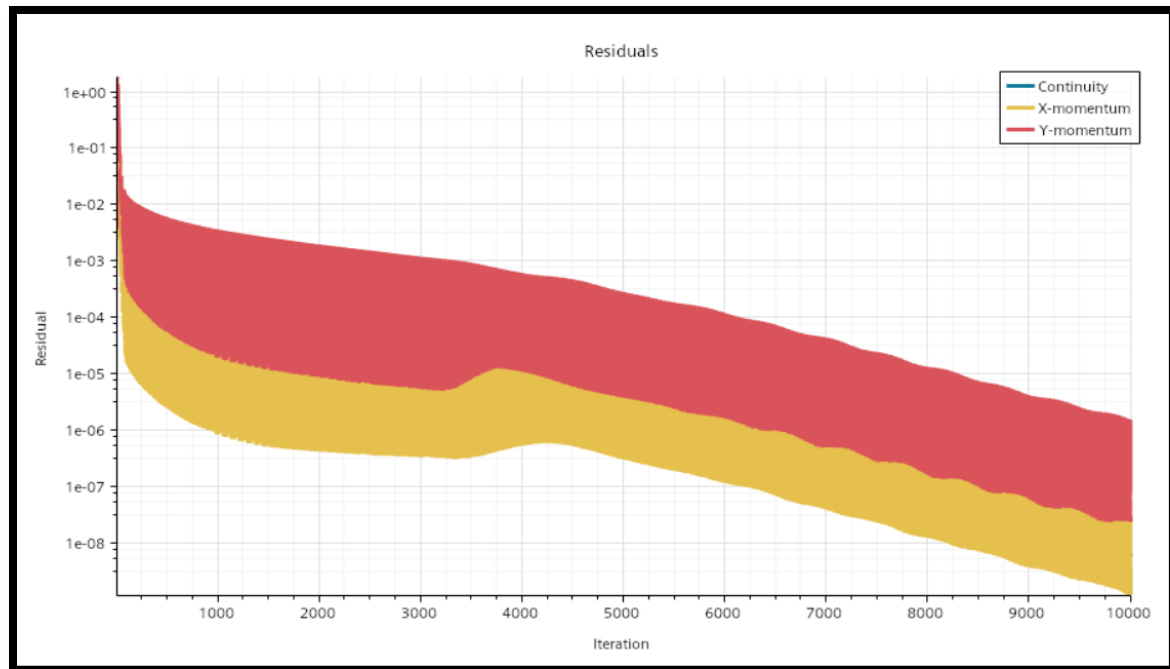
Appendix



Residual Plot of $Re=150$ simulation



Residual Plot of $Re=75$ simulation



Residual Plot of Re=25 simulation